

Exercise Sheet #8: AC Power and Resonances

Discussion: Dec 15, 2025

Exercise 1: ...

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Solution

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Exercise 2: ...

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Exercise 3: ...

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Exercise 4: ...

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Exercise 5: ...

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Exercise 6: Series Resonance

A series RLC network has $R = 2\text{ k}\Omega$, $L = 40\text{ mH}$, and $C = 1\text{ }\mu\text{F}$. Calculate the impedance at resonance and at one-fourth, one-half, twice, and four times the resonant frequency.

Solution

For a series RLC network with

$$R = 2\text{ k}\Omega, \quad L = 40\text{ mH}, \quad C = 1\text{ }\mu\text{F}$$

the series impedance is

$$\underline{Z}(j\omega) = R + j\left(\omega L - \frac{1}{\omega C}\right).$$

Resonance occurs when the net reactance is zero:

$$\omega_0 = \frac{1}{\sqrt{LC}}.$$

Substitute the values:

$$\omega_0 = \frac{1}{\sqrt{(40 \times 10^{-3})(1 \times 10^{-6})}} = 5000 \text{ rad s}^{-1}.$$

For a given angular frequency ω ,

$$X_L = \omega L, \quad X_C = \frac{1}{\omega C}, \quad X = X_L - X_C,$$

$$\underline{Z} = R + jX, \quad |\underline{Z}| = \sqrt{R^2 + X^2}, \quad \varphi = \arctan \frac{X}{R}.$$

We evaluate at $\omega = k\omega_0$ for $k = \frac{1}{4}, \frac{1}{2}, 1, 2, 4$.

k	ω (rad/s)	X_L (Ω)	X_C (Ω)	$\underline{Z}$ (k Ω)	$ \underline{Z} $ (k Ω)	φ
0.25	1250	50	800	$2 - j0.75$	2.136	-20.6°
0.5	2500	100	400	$2 - j0.3$	2.022	-8.5°
1	5000	200	200	2	2	0°
2	10000	400	100	$2 + j0.3$	2.022	$+8.5^\circ$
4	20000	800	50	$2 + j0.75$	2.136	$+20.6^\circ$

Observations:

- At resonance ($k = 1$) the inductive and capacitive reactances cancel, leaving purely resistive impedance $Z = R = 2000 \Omega$.
- The network is capacitive ($X < 0$) for frequencies below resonance and inductive ($X > 0$) for frequencies above resonance.
- The magnitudes at pairs $k = \frac{1}{4}$ and $k = 4$ are equal (symmetry), and likewise for $k = \frac{1}{2}$ and $k = 2$.

Exercise 7: Parallel Resonance

Rework Exercise 6 if the elements are connected in parallel.

Solution

For the parallel connection the total admittance is

$$\underline{Y}(j\omega) = \underbrace{\frac{1}{R}}_G + j \left(\underbrace{\omega C}_{B_C} + \underbrace{\left(-\frac{1}{\omega L}\right)}_{B_L} \right) = G + jB(\omega).$$

The impedance is the reciprocal of the admittance:

$$\underline{Z}(\omega) = \frac{1}{\underline{Y}(\omega)} = \frac{1}{G + jB} = \frac{G - jB}{G^2 + B^2}.$$

Its magnitude and phase are

$$|\underline{Z}| = \frac{1}{\sqrt{G^2 + B^2}}, \quad \varphi = \text{atan2}(\Im\{Z\}, \Re\{Z\}) = \arctan \frac{-B}{G}.$$

Parallel resonance (net susceptance zero) occurs at the same angular frequency as series resonance:

$$B(\omega_0) = 0 \quad \Rightarrow \quad \omega_0 = \frac{1}{\sqrt{LC}}.$$

With the given values

$$\omega_0 = \frac{1}{\sqrt{(40 \times 10^{-3})(1 \times 10^{-6})}} = 5000 \text{ rad s}^{-1}.$$

Also

$$G = \frac{1}{R} = \frac{1}{2000} = 5 \times 10^{-4} \text{ S}.$$

At resonance $B(\omega_0) = 0$, so

$$Y(\omega_0) = G \quad \Rightarrow \quad Z(\omega_0) = \frac{1}{G} = R = 2000 \Omega.$$

We compute for each ratio k :

$$\omega = k\omega_0, \quad B = \omega C - \frac{1}{\omega L}, \quad \underline{Z} = \frac{G - jB}{G^2 + B^2}.$$

k	ω (rad/s)	B_C (m Ω)	B_L (m Ω)	$\underline{Z}$ (Ω)	$ \underline{Z} $ (Ω)	φ
0.25	1250	1.25	20	$1.42 + j 53.30$	53	$+88.5^\circ$
0.5	2500	2.5	10	$8.85 + j 132.74$	133	$+86.2^\circ$
1	5000	5	5	2000	2000	0°
2	10000	10	2.5	$8.85 - j 132.74$	133	-86.2°
4	20000	20	1.25	$1.42 - j 53.30$	53	-88.5°

Notes on the results:

- At resonance the susceptances of the inductor and capacitor cancel and the parallel branch behaves as a pure conductance G , giving $Z(\omega_0) = R$.
- For frequencies below resonance ($k < 1$) the net susceptance B is negative (inductive behavior), producing an impedance with a large positive imaginary part (inductive reactance).
- For frequencies above resonance ($k > 1$) B is positive (capacitive behavior), producing impedance with a large negative imaginary part (capacitive reactance).
- The impedances occur in conjugate pairs for reciprocal frequency ratios (e.g. $k = 0.25$ and $k = 4$; $k = 0.5$ and $k = 2$).